Vivek Aruja

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Education

Western University London, Ontario
Bachelor of Engineering Science in Electrical Engineering and AI Systems Engineering *September 2023 – April 2028*

Experience

Business Analyst May 2025 – August 2025
Nanoleaf *Toronto, Ontario*

- Developed the frontend and backend of Nanoleaf's B2B site using Shopify to improve order efficiency and customer experience, resulting in a 15% increase in overall sales
- Optimized product listings by editing images and rewriting descriptions on Mirakl to ensure brand consistency and improved discoverability, leading to a 22% boost in customer engagement across Best Buy Canada and U.S. marketplaces
- Implemented automated workflows using Shopify Flow to notify internal teams via email when B2B orders were placed, improving operational responsiveness and reducing manual communication

Team Member May 2025 – Present
Sunstang Solar Car Team *London, Ontario*

- Attend weekly team meetings to collaborate with members on the development of a solar-powered vehicle
- Contributing to early-stage design and planning of the solar car, which will compete against other universities in international competitions
- Gaining exposure to interdisciplinary teamwork, sustainable energy technologies, and automotive engineering practices

Team Member September 2025 – Present
Western Capital Markets Club *London, Ontario*

- Participating in weekly workshops to deepen understanding of financial markets, investment analysis, and valuation techniques
- Collaborating with team members to evaluate equities, create mock portfolios, and present investment theses
- Developing foundational skills in financial modeling, Excel, and capital markets research through case studies and club resources

Undergraduate Research Assistant June 2024 – Sept 2024
Western University *London, Ontario*

- Developed a controller-based system using ROS to maneuver the Husky UGV, enabling precise control and navigation
- Created an Arduino-based door detection system with ultrasonic sensors and Arducam, enhancing environmental interaction capabilities
- Combined data from various sensors to improve the robot's situational awareness and decision-making processes

Technical Skills

Languages: Java, Python, C++, JavaScript, HTML/CSS, MATLAB
Frameworks: Arduino, Robot Operating System (ROS), React, Node.js, Tailwind CSS
Developer Tools: SOLIDWORKS, SketchUp, Onshape, IntelliJ, Visual Studio Code, Git, MicroCap, EAGLE
Operating Systems: Windows 10, Ubuntu, Linux
Skills: PCB Design, MS Office, Video Editing (Adobe Premiere Pro), Music Production (FL Studio), SwedaMart, Project Management, Warehouse Management

VIVEK ARUJA

ELECTRICAL & AI ENGINEERING AT THE UNIVERSITY OF WESTERN ONTARIO

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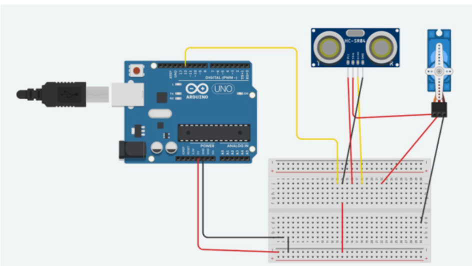
MOTION SENSOR GARBAGE CAN | PERSONAL PROJECT



What?

- Built a motion-sensor trashcan using an **Arduino**, **ultrasonic sensor**, and **servo motor**
- Designed to open the lid automatically when motion is detected

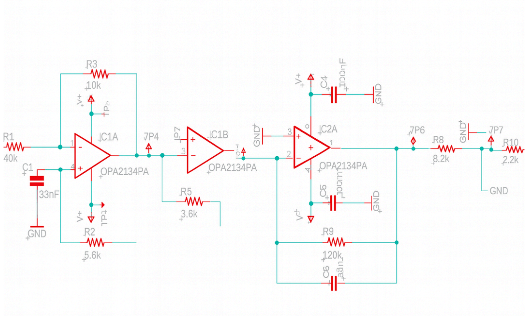
How?

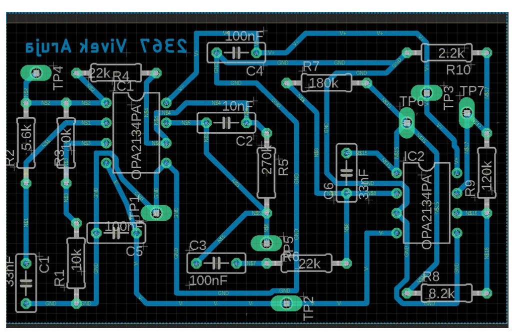
- Programmed and positioned electronic components for reliable operation
- Integrated servo motor control to lift the lid when a hand is waved in front of the sensor

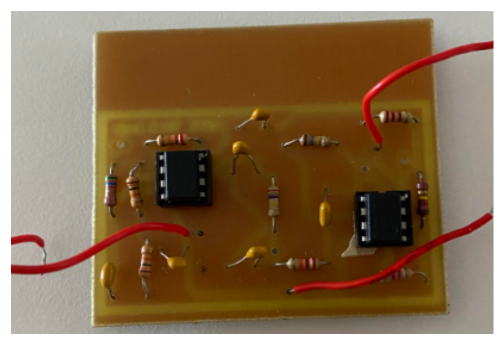
Results

- System operated smoothly and accurately, responding precisely to hand gestures

WAVEFORM GENERATOR | ECE 2240 PROJECT







What?

- Designed and built a **function generator PCB** capable of producing sine, square, and triangle waveforms
- Aimed to create a stable, soldered circuit rather than a breadboard prototype

How?

- Performed hand calculations for biasing and component values
- Simulated and verified circuit behavior in **EAGLE** software
- Designed PCB layout, soldered components, and developed troubleshooting and assembly plans

Results

- Successfully generated all three waveforms with stable output
- Gained hands-on experience in PCB design, soldering, and circuit testing